



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)



Project Synopsis

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Project Title: IntelliTwin: AI-Powered Digital Twin Platform for Smart Building Monitoring, Predictive Maintenance and Intelligent Energy Optimization

Introduction:

A Digital Twin is a virtual, continuously updated representation of a physical asset or system that mirrors its real-time state through streaming data. Originally developed for aerospace and manufacturing, the concept has extended into smart buildings, where digital twins now monitor operating conditions, anticipate equipment degradation, and optimize resource consumption across a building's lifecycle.

Smart buildings need digital twins because occupancy, equipment load, and environmental conditions change continuously and interact in non-linear ways — a virtual model lets facility staff test decisions before committing to them physically. Existing monitoring systems, however, are largely limited to displaying live sensor values without correlating them to failure risk, energy trends, or operational efficiency. Maintenance stays reactive or schedule-based, and energy inefficiencies often go unnoticed until they show up on a utility bill.

Artificial intelligence and predictive analytics close this gap by learning patterns from historical operating data and turning them into forward-looking insight — for instance, the probability that a compressor will fail within a given window, or the expected impact of an occupancy surge on power draw. This project, IntelliTwin, proposes a software-based Digital Twin platform for smart buildings that operates entirely without physical IoT hardware. A Virtual IoT Sensor Simulator generates realistic sensor data conditioned on occupancy, weather, equipment usage, and time of day. This data is processed by a full-stack web application and analyzed through machine learning models for predictive maintenance, anomaly detection, and energy optimization, then rendered through an interactive 3D dashboard — giving facility managers the intelligence of a hardware-based smart building system without the deployment cost or complexity.

Objectives:

- Design a Virtual IoT Sensor Simulator that generates realistic, behaviourally-consistent sensor data without physical hardware
- Develop a full-stack platform (React, Node.js, MongoDB) to ingest, store, and serve real-time simulated sensor data
- Build a Random Forest-based predictive maintenance model that estimates equipment failure probability
- Develop an AI-based energy consumption forecasting module to support intelligent energy optimization.
- Develop an Isolation Forest-based anomaly detection module for real-time abnormal readings
- Design an interactive 3D Digital Twin dashboard (Three.js) for building/floor/room-level visualization

Software Requirements –

- Frontend: React.js
- Styling: Tailwind CSS
- 3D Visualization: Three.js
- Charts/Graphs: Chart.js
- Backend: Node.js, Express.js
- Authentication: JWT
- Database: MongoDB
- AI/ML: Python, Scikit-learn
- Data Handling: Pandas, NumPy
- Deployment (optional): Docker, AWS/Render

Hardware Requirements –

- **No physical IoT sensors, microcontrollers, or field hardware required** — the project uses a software-based Virtual IoT Sensor Simulator instead
- Development laptop: minimum 8 GB RAM, Intel i5 processor (or equivalent)
- At least 10 GB free disk space for dependencies, datasets, and trained models
- Stable internet connection for package installation and optional cloud deployment

Abstract:

Buildings account for a significant share of global energy consumption, yet most building management systems expose only the current state of the environment without any capacity to anticipate failures or recommend corrective action. This forces facility teams into a reactive posture: equipment is serviced only after it degrades, and energy waste persists undetected until

it appears in operating costs. Deploying physical IoT sensor networks to solve this is expensive and slow, limiting the ability of students and early-stage teams to prototype intelligent building systems.

IntelliTwin is built on the premise that the intelligence layer of a smart building — predictive maintenance, anomaly detection, and energy optimization — can be developed and validated independently of physical sensor hardware, provided the simulated data is behaviourally realistic. A Virtual IoT Sensor Simulator generates continuous readings for temperature, humidity, occupancy, power, smoke, and motion, driven by rules that couple occupancy, weather, equipment usage, and time of day to their expected physical effects.

The pipeline follows a layered architecture: the simulator publishes data through REST APIs to a Node.js/Express backend, which persists records in MongoDB; a Python-based AI engine then consumes this data to generate predictions. Four AI components form the core: a Random Forest classifier for predictive maintenance (estimating failure probability from temperature, power, runtime, and equipment age); a regression model for short-term energy forecasting; an Isolation Forest model for unsupervised anomaly detection across sensor streams; and a rule-based recommendation engine that converts model outputs into actionable suggestions.

Major modules include role-based authentication (Admin, Building Manager, Technician), the Virtual Sensor Simulator, a real-time monitoring dashboard, an interactive 3D Digital Twin, a what-if scenario simulation module, alerts/notifications, and automated PDF reporting.

The expected outcome is a working prototype demonstrating earlier detection of equipment degradation, quantified energy-saving recommendations, and an intuitive spatial interface for non-technical staff. Because the architecture communicates through standard REST interfaces, physical IoT sensors could replace the simulator in future work without redesigning the backend, AI engine, or frontend. Real-world applications extend to educational campuses, offices, hospitals, and residential complexes, where predictive, AI-driven building management can reduce operating costs, extend equipment lifespan, and support Net Zero energy goals.

Literature Survey

Paper 1

Title: *A Comprehensive Review of Digital Twin Technology in Building Energy Consumption Forecasting*

IEEE Reference:

[1] M. Boukaf, F. Fadli, and N. Meskin, "A Comprehensive Review of Digital Twin Technology in Building Energy Consumption Forecasting," *IEEE Access*, vol. 12, pp. 187778–187799, 2024, doi: 10.1109/ACCESS.2024.3498107.

Summary:

This paper reviews Digital Twin technology for building energy forecasting using IoT, Artificial Intelligence, and Machine Learning. It discusses real-time monitoring, predictive maintenance, and energy optimization techniques for smart buildings.

Research Gap:

The study is primarily a review and does not provide a complete software platform integrating Digital Twin visualization, AI prediction, and scenario simulation.

Paper 2

Title: *A Review of Building Digital Twins to Improve Energy Efficiency in the Building Operational Stage*

IEEE Reference:

[2] A. S. Cespedes-Cubides and M. Jradi, "A Review of Building Digital Twins to Improve Energy Efficiency in the Building Operational Stage," *Energy Informatics*, vol. 7, no. 1, 2024, doi: 10.1186/s42162-024-00313-7.

Summary:

The paper reviews the application of Digital Twin technology for improving building energy efficiency through occupancy analysis, HVAC optimization, and real-time monitoring.

Research Gap:

The work focuses on operational energy efficiency and does not integrate predictive maintenance, virtual sensor simulation, or intelligent recommendation systems.

Paper 3

Title: *Digital Twins for Reducing Energy Consumption in Buildings: A Review*

IEEE Reference:

[3] B. Arsecularatne, N. Rodrigo, and R. Chang, "Digital Twins for Reducing Energy Consumption in Buildings: A Review," *Sustainability*, vol. 16, no. 21, Art. no. 9275, 2024, doi: 10.3390/su16219275.

Summary:

This paper reviews Digital Twin applications for reducing energy consumption in smart buildings using occupancy analysis, real-time monitoring, and AI-based energy management.

Research Gap:

The paper does not present an integrated software solution combining predictive maintenance, Digital Twin visualization, and what-if scenario analysis.

Paper 4

Title: *Ontology-enabled AI Agent-driven Intelligent Digital Twins for Building Operations and Maintenance*

IEEE Reference:

[4] S. Yoon, J. Song, and J. Li, "Ontology-enabled AI Agent-driven Intelligent Digital Twins for Building Operations and Maintenance," *Journal of Building Engineering*, vol. 108, Art. no. 112802, Aug. 2025, doi: 10.1016/j.job.2025.112802.

Summary:

This paper proposes an AI agent-driven intelligent Digital Twin framework for building operations and maintenance. It integrates ontology-based knowledge representation with AI agents to support predictive maintenance, fault detection, and intelligent decision-making. The framework demonstrates improved operational efficiency through Digital Twin-enabled building management.

Research Gap:

The study focuses on AI-assisted building operation and maintenance but does not incorporate a software-based Virtual IoT Sensor Simulator or interactive what-if scenario analysis for intelligent energy optimization.

Paper 5

Title: *Energy Digital Twin Technology for Industrial Energy Management: Classification, Challenges and Future*

IEEE Reference:

[5] W. Yu, P. Patros, B. Young, E. Klinac, and T. G. Walmsley, "Energy Digital Twin Technology for Industrial Energy Management: Classification, Challenges and Future," *Renewable and Sustainable Energy Reviews*, vol. 161, Art. no. 112407, 2022.

Summary:

This paper reviews Digital Twin technologies for energy management, focusing on AI, IoT, and data analytics for improving system efficiency and sustainability.

Research Gap:

The research mainly addresses industrial energy systems and does not provide a software-based Digital Twin framework specifically for smart building monitoring.

Existing System & its Drawback:

Existing System

- Most current BMS and IoT dashboards are limited to real-time data acquisition and visualization
- Display temperature, humidity, occupancy, and power readings without correlating them to failure risk or efficiency
- Maintenance is either scheduled at fixed calendar intervals regardless of actual equipment condition, or triggered only after a visible fault occurs
- Alerting is typically threshold-based, flagging values only once they cross a static limit
- Nearly all deployed implementations are hardware-dependent, requiring procurement, installation, and upkeep of physical sensors and gateways

Drawbacks

- No predictive capability — failures are caught only after occurring
- No energy forecasting or optimization recommendations
- No way to simulate future or hypothetical scenarios
- Heavy reliance on costly physical IoT hardware
- Frequent false positives/negatives from threshold-based alerts
- Limited or purely tabular visualization unsuited to multi-floor buildings
- No role-differentiated workflows for different types of users
- Manual, time-consuming reporting processes

Proposed System & Advantages:

Proposed System

- Architecture flow: Virtual Sensor Simulator → REST API → Node.js Backend → MongoDB → Python AI Engine → React Dashboard → 3D Digital Twin
- The simulator generates behaviourally-consistent readings per room, coupling occupancy to power/temperature and weather to cooling load
- Readings flow via REST APIs to the Node.js/Express backend, which persists data in MongoDB for both real-time queries and AI training
- The Python AI Engine runs predictive maintenance, energy forecasting, anomaly detection, and recommendation models, exposing results back to the backend
- The React dashboard renders live graphs and alerts
- The Three.js 3D Digital Twin lets users navigate the building visually, with room colour/state reflecting health and risk
- The what-if module lets Admins/Managers test hypothetical conditions before real-world decisions
- Role-based access differentiates Admin, Manager, and Technician views

Advantages

- Eliminates hardware cost and installation complexity
- Enables proactive maintenance via early failure prediction
- Supports proactive energy optimization instead of after-the-fact analysis
- Improves alert accuracy via ML-based anomaly detection over static thresholds
- Allows risk-free scenario testing before real-world implementation
- Provides an intuitive 3D interface easier to interpret than tabular dashboards
- Differentiates functionality by role for better usability and accountability
- Automates reporting, saving staff time and enabling faster decisions
- Modular REST-based architecture that can accept real IoT hardware later without redesign
- Low cost and fast deployment, making it well suited for academic prototyping

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